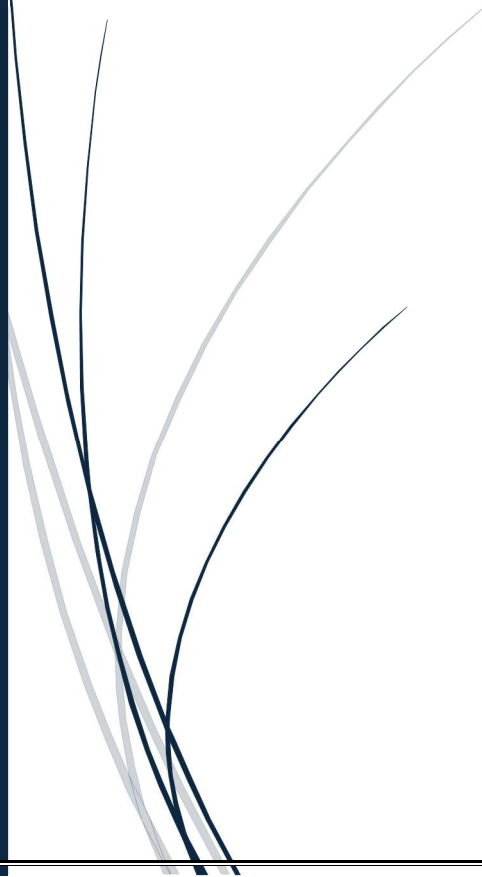




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Assignment 1

Advanced Data Visualization



Mihir Mistry
NF1029056

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1. Project Overview:

The goal of this project was to conduct a structured exploratory data analysis and business intelligence analysis using Tableau. The project was centered around bringing together the various datasets, addressing imperfect data, creating analytical visualizations, discovering patterns that make profitable business and drawing business insights with interactive dashboards.

The database was made up of two worksheets:

- Orders
- Details

The project required:

- Data integration and relationship modeling
- Addressing data quality problems
- Calculating analytically created fields
- Performing exploratory analysis,
- Recognizing losses,
- Developing advanced Tableau dashboards for executive and operational intelligence.

Multiple analytical worksheets have been added to the final solution, along with 3 integrated dashboards:

- Executive Sales & Profitability Dashboard
- A dashboard displaying Predictive Analytics & Portfolio Intelligence.
- Operational Risk & Profitability Diagnostics Dashboard

2. Data Modeling Decisions:

The data was spread between two tables that had a relationship between them and was ordered by the transaction number. The Orders table mainly comprised the following attributes:

- Order ID
- Order Date
- Region
- Customer ID
- Category
- Sub-Category

The Details table had numerical business measures such as:

- Sales
- Profit
- Quantity

A connection (relationship) was built between the two tables using the Order ID field in order to properly join the datasets.

A one to many relationship was found and implemented as it was:

- There may be several transaction records on the product level within one order,
- Each detail record is associated with a single order.

Such a relation structure enabled to prevent any error of duplication and maintained the accuracy of aggregation during the analysis.

The data model was well-designed to assure that:

- proper aggregation behavior,
- Accurate sales and profit calculations,
- and Trusted dashboard analytics.

The left relationship structure was successfully implemented via Tableau's relational model to keep all order records when there are inconsistencies at the detail level.

The correct data types were also checked:

- To ensure the dates could be used in calculations, they were stored as Date data types
- numeric values were assigned as numeric points
- and categorical fields were left as dimensions

3. Data Cleaning & Preparation Approach

The data set was examined for any data quality problems prior to analysis such as:

- missing values
- inconsistent formatting
- aggregation problems
- and possible duplication problems

The following preparation steps were carried out:

- Confirmation of data which is missing or incomplete
- correction of data types
- validation of Order ID relationships
- Consistency and category checks across regions.

Also developed calculated fields to enhance analytical depth and business intelligence capability.

Key Calculated Fields

Profit Ratio

- Measures the profitability efficiency against the level of sales activity.
- This is the profit ratio. This is the profit ratio: Profit/Sales.
- This computation identified high and low efficiency product segments.

Sales per Quantity

- Measures the amount of revenue that can be collected from each unit sold.
- Sales per Quantity = Sales / Quantity
- This enabled the pricing and performance of the operation to be assessed.

Order-Level Profit

- A calculated field to measure total profitability at the order level using aggregation.
- This prevented the aggregation distortion and enhanced transaction level diagnostics.

Loss Severity Classification

Use to categorize transactions as:

- High Loss
- Moderate Loss
- No Loss

This classification was used for the negative profit diagnostics and operational risk analysis.

4. Exploratory Analysis & Dashboard Development

Extensive exploratory data analysis using a variety of Tableau visualizations was conducted as part of this project.

The following analytical views were designed:

- Monthly Sales Trend Analysis
- Profit Distribution Analysis
- Regional Performance Comparison
- Profitability Analysis – Category vs Sub-Category.
- Analyzing the relationship between sales and profit.
- Forecasting & Prediction: Trend Analysis
- Negative Profit Diagnostics
- Heat Map Analysis
- Tree Map Portfolio Analysis
- Histogram Distribution Analysis
- Top-N Product Contribution Analysis is a tool that measures the contribution of each product.
- This is the Dynamic Cascading Filter Dashboard.

Advanced Tableau analytical features were added, such as:

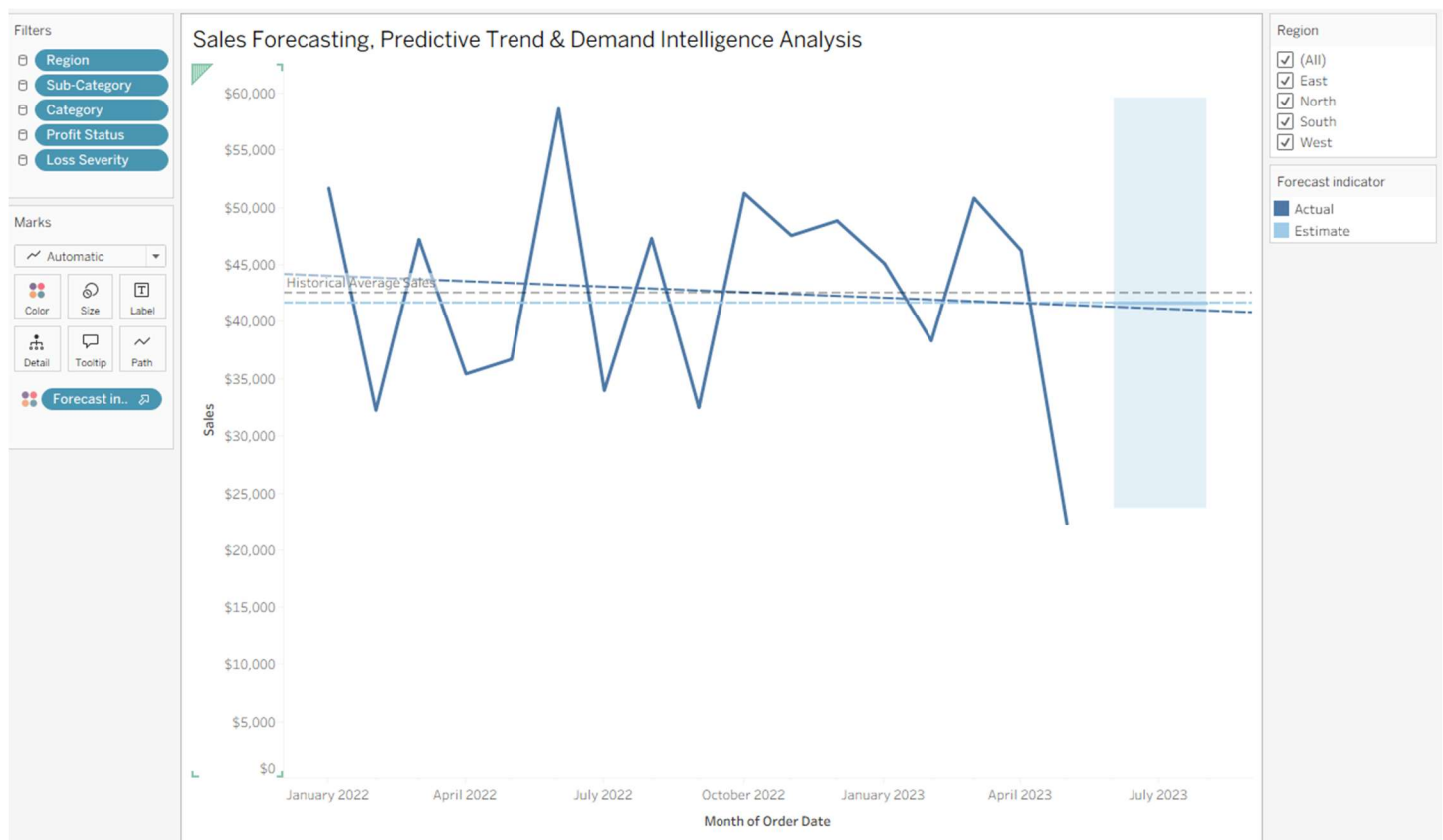
- Forecasting
- Reference Lines
- Moving Averages
- Trend Analysis
- Heat Maps
- Treemaps
- Histogram Binning
- Dynamic Filters
- Parameter-Based Risk Thresholds
- Quadrant Analysis

The dashboards were designed to help both:

- Executive-level decision-making,
- And Operational diagnostic analysis.

5. Key Insights

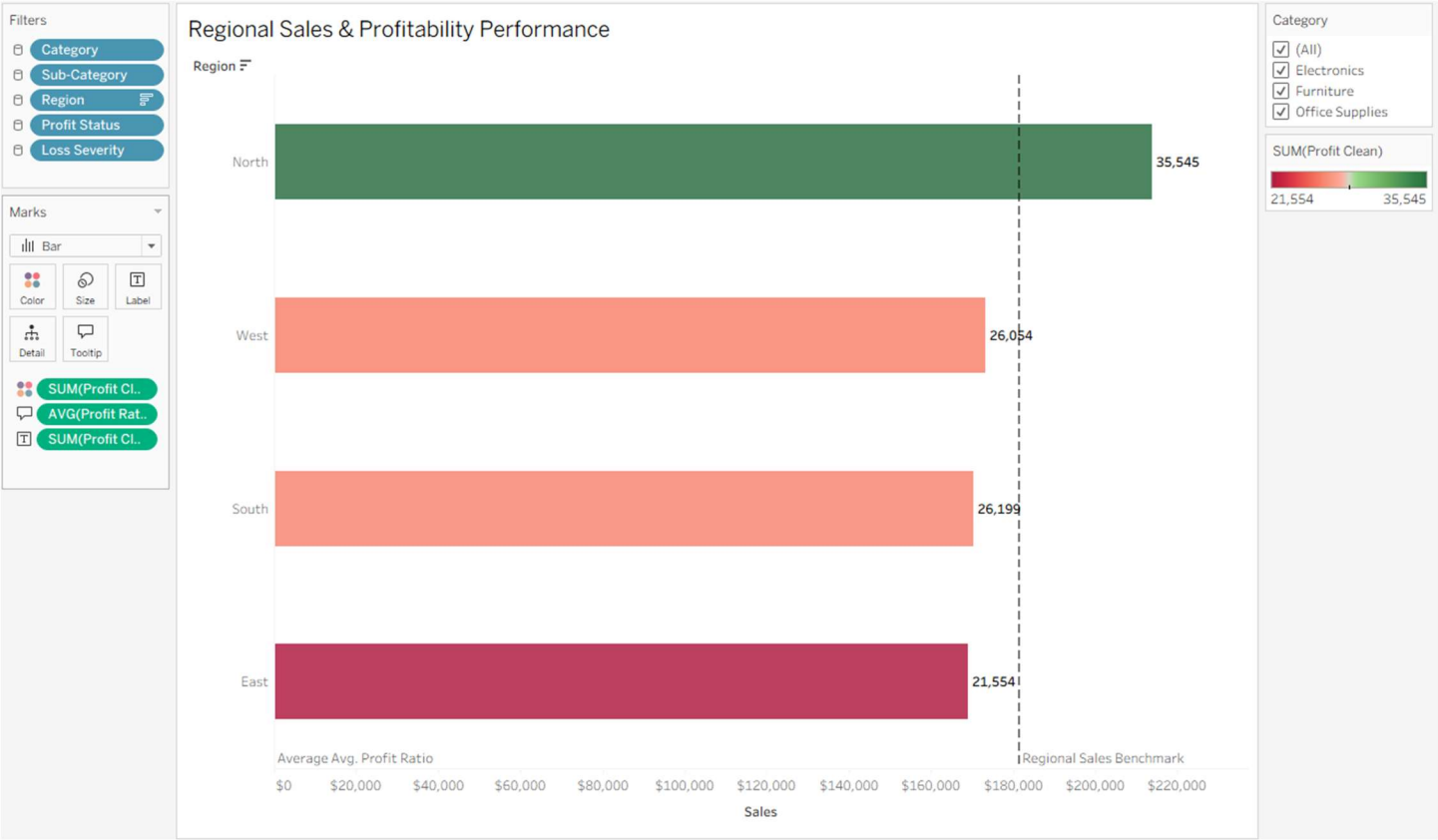
Insight 1 – Regional Profitability Concentration



Overall, the sales and profitability numbers of the North region were the highest in comparison to the other regions. This indicates a more efficient operation of that market and concentration of customer demand.

The profitability performance in East and South regions was relatively weak, suggesting opportunities for pricing optimization and operational improvements.

Insight 2: Profitability is not directly correlated to Sales.

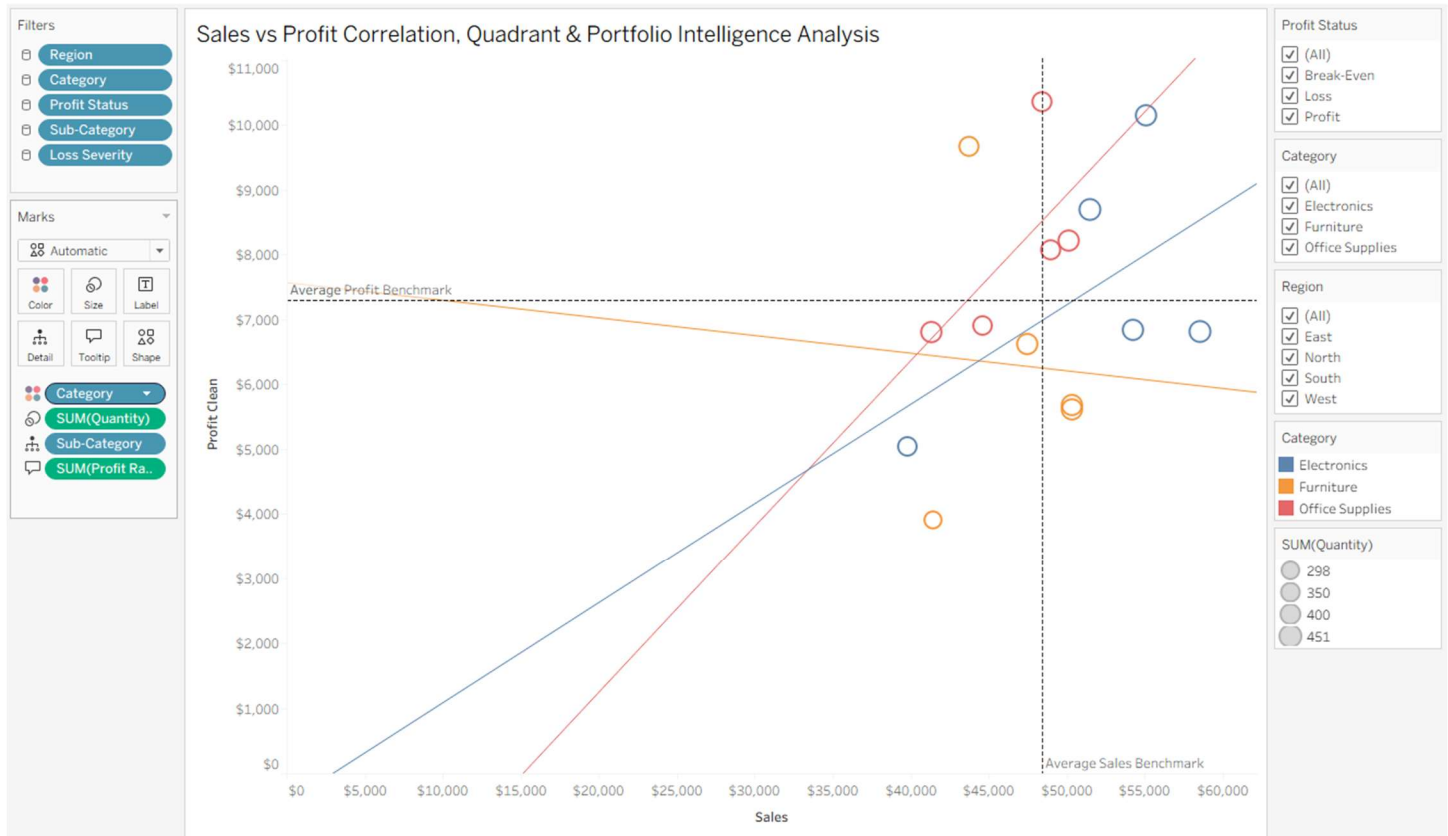


Sales vs Profit scatter plot analysis showed that the amount of sales made does not always lead to increased profit.

There were some products with high revenue and low efficiency, and a few products with medium sales and relatively high efficiency.

This is an insight that shows that there is a need to balance revenue growth with profitability optimization.

Insight 3: Understanding how negative patterns of profit arise is systematic.



The negative profit dashboard showed that losses were focused on certain:

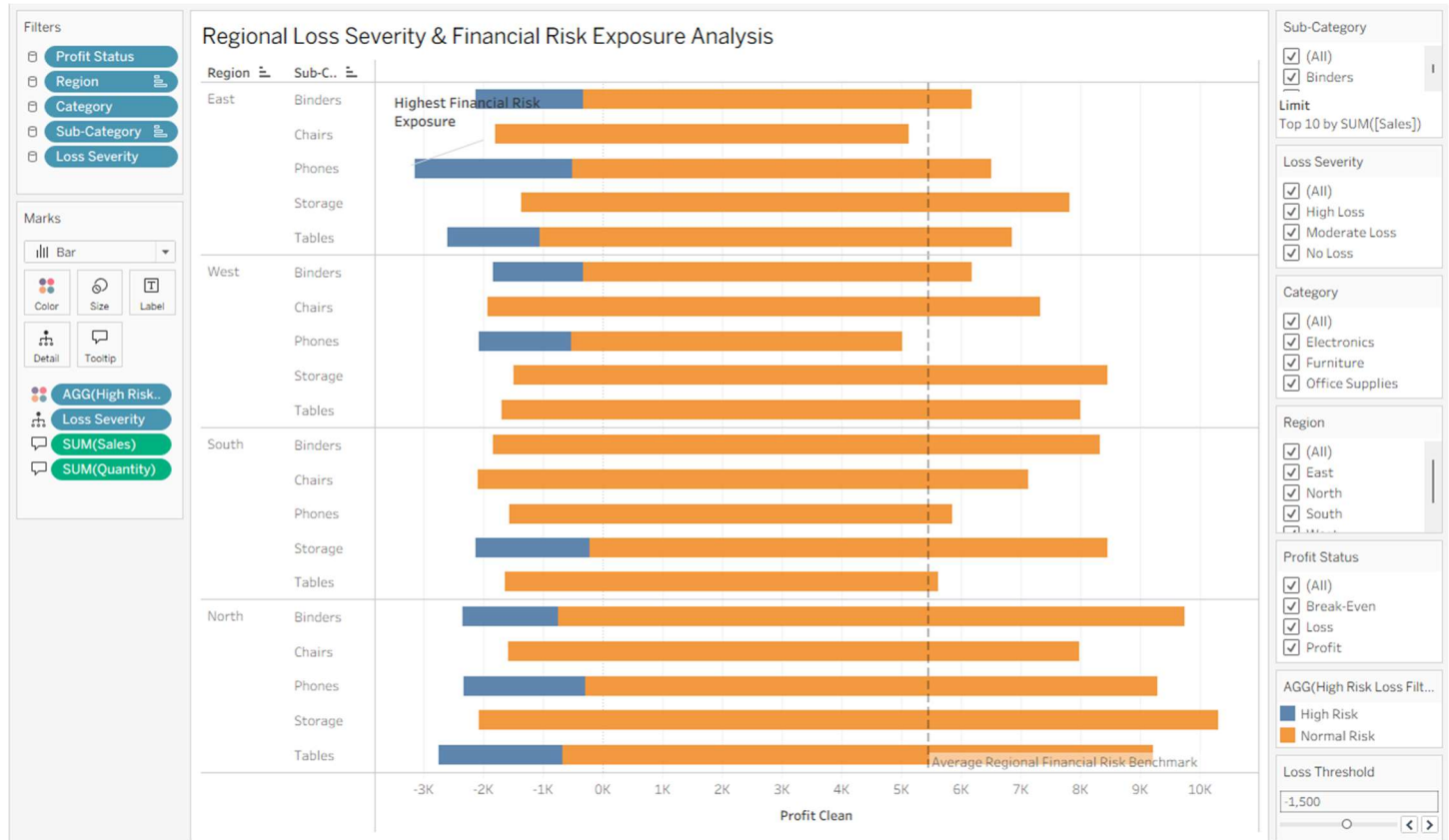
- product categories
- sub-categories
- and regions.

Losses were not uniform.

Selected Furniture and Phone segments may be experiencing a repetitive trend of negative profitability, which might be due to:

- aggressive discounting,
- high operational costs,
- inefficient shipping,
- or flaws in pricing strategies.

Insight 4 — Portfolio Concentration Exists: This insight identifies if there is a portfolio concentration.

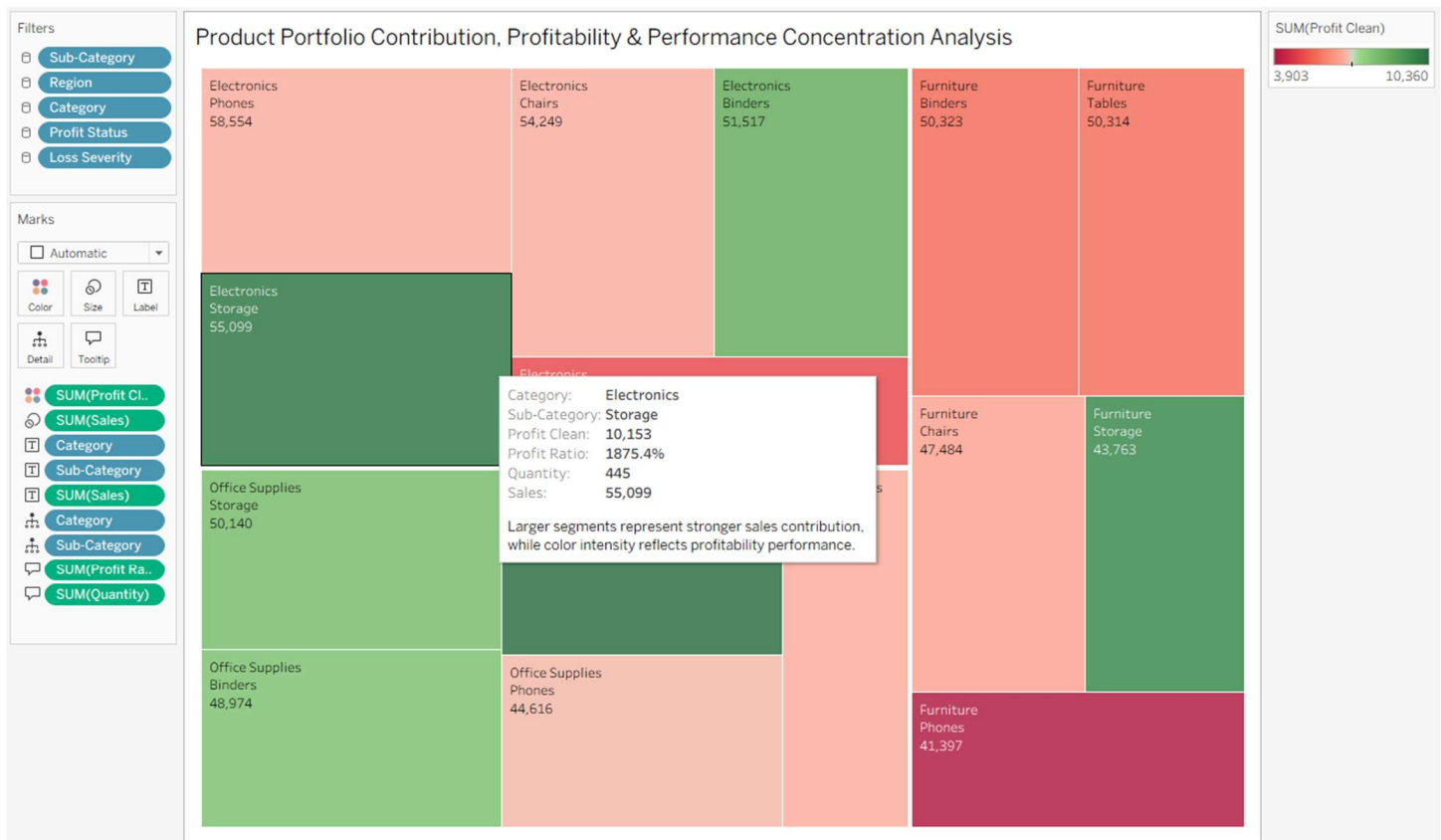


Portfolio analysis on the tree map and heat map showed high concentration of the portfolio in the following sub-categories:

- Storage,
- Binders,
- Tables.

This means that few product segments play a significant role in business performance.

Insight 5 - Forecasting indicates stable but volatile demand



The overall sales were relatively stable over time as suggested by the analysis of the forecasts, but with fluctuations and confidence intervals there was moderate seasonal volatility.

This implies that:

- inventory planning,
- demand forecasting,
- and promotional optimization
- Stability of business performance is important and continues to be so.

6. Challenges Encountered:

There were a few obstacles and issues faced during this project.

A key difficulty was getting multiple tables to aggregate together without losing their ability to be aggregated. Join mismatches may easily lead to duplication problems and misrepresent or skew profitability figures.

Another challenge was to create dashboards that struck a balance between:

- analytical depth,
- visual clarity,
- and business storytelling.

Advanced visualizations like risk diagnostics, quadrant analysis and forecasts needed careful formatting and interpretation.

Optimizing dashboard layout and understanding how to interact with the filters was also a major challenge in terms of usability and readability.

Even in the face of these obstacles, the final dashboards seamlessly merged cutting-edge analytics, predictive intelligence and operational diagnostics into a professional Tableau business intelligence solution.

7. Conclusion:

This project showcases the power of Tableau as more than just a tool for visualization to support:

- Advanced exploratory analysis,
- Predictive analytics,
- Profitability diagnostics,
- Executive business intelligence.

The project delivers actionable insights through structured data modeling, development of calculated fields, integration with dashboards, and interpretation of the analytical data.

- Sales trends,
- Regional performance,
- Product profitability,
- Operational risks,
- Portfolio optimization opportunities.

The final solution is analytical, data-driven, and a storyteller of business intelligence, for strategic decision making.

8. AI Disclosure:

In this project, AI tools, such as OpenAI ChatGPT, were employed in a limited manner to provide support.

The main use of AI assistance was:

- Basic tips for formatting and organizing the layout of dashboards, and
- Suggestions on the selection of color palette and dashboard aesthetics,
- Suggestions on how to organize the presentation and write a report,
- Assist the user in finding features and formatting on Tableau during the development of a dashboard.

The following analytical aspects of the project were done independently

- data integration and relationship modelling
- For example, Tableau Prep data preparation
- Dealing with missing data and aggregation logic
- Use of calculated fields
- dashboard development
- visualization selection
- analytical interpretation
- business insight generation
- Final decision-making during the project.

All Tableau visualizations and dashboards submitted were created, executed, validated and interpreted independently.